Q. Find minimum spanning tree G, its weight of graph using Kruskal's Algorithm

Step 1 Initialize MST with all vertices, but no edges

Step 2 Pick edge : (2,7) No cycle is formed So, include it

Step 3 Pick edge : (2,3) No cycle is formed So, include it

Step 4 Pick edge : (2,8) No cycle is formed So, include it

Step 5 Pick edge : (3,4) No cycle So, include it

Step 6 Pick edge : (7,4) But cycle is formed So, discard it

Step 7 Pick edge : (5,4) No cycle So, include it

Step 8 Pick edge (5,7) cycle is formed So, discard it

Step 9 Pick edge : (4,6) No cycle So, include it

* Prim's Algorithm

1) Choose any vertex of graph $G(V,E)$
2) Choose an edge (say $e_1 = (v_1, v_2)$) such that $(v_1 \neq v_2)$ and e_1 has the smallest cost among the edges of G incident with v_1.
3) If edges $e_1, e_2, \ldots e_i$ have been chosen, forming vertices $v_1, v_2, \ldots v_i, v_{i+1}$ and $v_k \notin (v_1, v_2, \ldots v_i, v_{i+1})$ and e_{i+1} has the smallest weight among all the edges of G incident with $(v_1, v_2, \ldots v_i, v_{i+1})$
4) Stop after $(n-1)$ edges have been chosen otherwise go to step 3

Here, no of vertices, $n = 7$
no of edges included $= n-1 = 7-1 = 6$
So, stop here.
Total cost/weight $= 9+7+3+8+10+4$
$= 64$

Q. Construct MST for following graph using prim's algorithm

Step 4: Edges incident on V_1, V_2, V_3 with their weights :

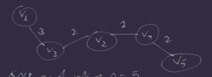
$W(V_1, V_2) = 4$ $W(V_2, V_4) = 1$ ✓
$W(V_1, V_4) = 3$ $W(V_2, V_5) = 3$
$W(V_1, V_5) = 3$ $W(V_3, V_4) = 3$

Choose (V_2, V_4) edge as it has minimum weight

Step 5: Edges incident on V_1, V_2, V_3, V_4 with their weight :

$W(V_1, V_3) = 4$ $W(V_2, V_5) = 3$
$W(V_1, V_4) = 3$ $W(V_3, V_4) = 3$
$W(V_3, V_5) = 3$ $W(V_4, V_5) = 2$ ✓

Choose (V_4, V_5) as it has minimum weight



Since, no of vertices, $n = 5$
no of edges selected $= n-1 = 5-1 = 4$
So, stop here
Total cost/weight $= 3+2+1+2$
$= 8$

Q. Using Both algorithm, draw MST;

Q. Find MST for following

Step 1 Remove all self loops from graph

Step 2 Remove parallel edges which have maximum weight, leaving only one edge with minimum weight

Now use Kruskal's or prim's algorithm to find MST

Imp

Shortest path algorithm :

(Dijkstra's Algorithm)

→ Greedy algorithm approach